

Master Thesis

Fiete Scheel

Automated Translation into Simple German

Language as a barrier: Complex texts exclude people from information and social participation.

High effort: To date, the creation of simple german texts has been almost exclusively manual and time-consuming.

Research gap:

Lack of robust, data-driven methods for distinguishing between standard and simple language in German.

Lack of automated metrics for the qualitative assessment of simplifications.

Components of this work

Week 1

Data Research

Problem:

It is often very difficult to find exact parallel texts (such as news articles or blog posts) on the internet that are written in standard language (or "complex language") and simple language, and that can be systematically matched with one another.

sozialpolitik.com (Sloution with URL)

Here, the blog posts have the same title in both language versions.

The "Easy Language" version is easy to identify and access, as the URL simply has the **prefix** "Is-" added to the beginning.

While the URLs for the blog posts differ, there is a **button** on each page that allows you to **switch between "Easy Language" and "Standard Language."**

sozialpolitik.com (normal german)

sozialpolitik.com (simple german)

At Mitteldeutscher Rundfunk, the links aren't embedded in the URL, but the user guidance within the text is excellent.

Below every article in Easy Language, there is a **direct link** to its counterpart in standard language and **vice versa**.

Not every article in standard language is translated into Easy Language, but **every article in Easy Language is translated into standard language** (as verified by random sampling).

mdr.de (normal german)

mdr.de (simple german)

Unfortunately, Tagesschau stands out negatively when it comes to integrating standard and simplified language, as it lacks a user-friendly and systematic structure:

- **Lack of links:** There is no button or direct link within the articles to easily switch back and forth between the different language levels.
- **Inconsistent URL structure:** While the basic structure of the URLs appears well-thought-out, in practice the articles are named inconsistently. There is no reliable pattern (such as a fixed prefix), making a systematic or even automated mapping of the texts impossible.
- **High manual effort:** To find parallel texts, the only option is a time-consuming manual search for the semantically matching counterpart.
- **Significant reduction in content:** Even if you have found the corresponding pair of articles, a direct text comparison is difficult because the version in Easy Language has been drastically shortened and greatly simplified in terms of content.

Next Steps

Researching additional sources: Searching for other news portals or official government websites (e.g., bpb.de) that systematically provide LS texts.

Quantification of sources: Determining the approximate number of available article pairs on mdr.de and sozialpolitik.com .

Scraper development: Creating a prototype for the automated extraction of articles from sozialpolitik.com (using URL logic).

MDR crawler: Investigation into whether the links between LS and standard language on mdr.de can be systematically crawled.

Matching strategy for Tagesschau: Evaluation of whether semantic similarity analyses (e.g., via embeddings) can be used to automatically establish the missing links in Tagesschau.

Week 2-3

Evaluation of institutional and governmental websites.

Observations: Significant availability of "Easy Language" (Leichte Sprache) sections, but alignment Issue.

Alignment Issue: Most institutional pages provide summaries or simplified overviews rather than 1:1 translations of standard articles.

Terminological Inconsistency: Frequent overlap and confusion between "Plain Language" (Einfache Sprache) and "Easy Language."

Saarländischer Rundfunk (SR)

Problem: Standard language content is often video/audio (broadcast format), while the Easy Language version is a brief text summary.

Impact: Automated 1:1 alignment is difficult due to the format mismatch.

Technical Workaround: Using metadata (Open Graph/OG-Images) to verify if two different formats refer to the same news event + Date of Publishment

Media Portals with High Alignment Potential

MDR (Mitteldeutscher Rundfunk): High suitability for scraping due to clear XML sitemaps and explicit "linking mechanisms" in the HTML structure.

taz: Alignment is present but less systematic; links are often embedded manually in italicized paragraphs at the end of articles with word "hier" .

Conclusion: Media portals show significantly higher potential for 1:1 alignment than administrative/government portals.

Terminological Anchors: Dictionaries



Image

Objective: Establishing a "Ground Truth" for specific terminology.

Resources Analyzed:

- **Hurraki:** A specialized Wiki for Easy Language.
- **Nachrichtenleicht:** A dictionary section providing definitions for complex terms.

Benefit: These provide direct 1:1 word/phrase mappings, essential for terminological consistency.

Literature Review: Existing Corpora (Status 2023)

Analysis of Klaper et al. (2013)

Pros: High-quality 1:1 alignment; approx. 70,000 tokens.

Cons: "Source decay"---many URLs from 2013 are no longer active.

Next Step: Evaluating the reconstruction of this corpus using the Wayback Machine (Internet Archive).

Literature Review: Existing Corpora (Status 2023)

A New Aligned Simple German Corpus (ACL 2023)

Analysis: Investigating provided crawler scripts and alignment logic.

Key Insight: The project moves away from generic scraping toward source-specific extraction heuristics to ensure high-quality sentence alignment.

Extraction Strategies

Apotheken Umschau: Targeting specific link titles

`title="hier"` for back-references.

Behindertenbeauftragter: Identifying language switches via CSS class patterns `.c-language-switch`.

Sozialpolitik.com: Filtering for links with the class `underline` `easy` and specific `hreflang` attributes.

```
<a href="/es/die-arbeits-welt" hreflang="de-DE" class="underline easy">
  Leichte Sprache
</a>
```

Extraction Strategies

Brand Eins: Unique case where both languages exist on the same URL; differentiation achieved through CSS color coding (Red text = Easy Language).

Stadt Köln: Using exact string matching for navigational links ("Diese Seite in Alltags-Sprache lesen").

Summary: Technical research confirms a "one-size-fits-all" scraper is ineffective; custom logic for each domain is required.

Conclusion & Next Steps

Current Activity: Development of a quantification script to scan identified sources and calculate current token counts.

Pipeline Integration: These counting scripts serve as "pre-scrapers" to validate data density before full extraction.

Literature & Expansion: Systematic review of remaining papers from the "Existing Corpora (Status 2023)" list.

Identification of additional sources and site-specific scraping mechanisms.

Goal: Adaptation of the Toborek et al. crawler logic to the newly identified structures (MDR, taz, etc.).

Week 4

Initial Scraping for Token Count

Goal: Create a diverse, multi-domain corpus exceeding current benchmarks in quality and thematic breadth.

The Pipeline:

Discovery: Crawling target websites for "Simple Language" (LS) sections.

Alignment: Matching LS articles to their "Standard Language" (AS) originals.

Extraction: Pulling raw content for initial analysis.

Target Scope: News, Health, Administration, Social Policy, and Economy.

Results: Media & News

MDR (Mitteldeutscher Rundfunk):

298 LS articles scanned → 242 successfully aligned.

Largest news source in the corpus (~300k tokens total).

taz (die tageszeitung):

Small but high-quality subset (7 pairs aligned).

Challenge: One LS text often summarizes multiple AS articles.

Brand Eins:

90 articles scanned → 34 pairs aligned.

Focus on economic and social narratives.

Results: Public Sector & Administration

Hamburg.de:

156 articles scanned → 155 initial pairs.

Broad coverage of city services and regulations.

Stadt Köln:

95 articles scanned → 57 pairs aligned.

Historical data recovered from archives.

Behindertenbeauftragter:

95 articles scanned → 73 pairs aligned.

Policy-focused content with technical terminology.

Results: Specialized Topics

Apotheken Umschau (Health/Medical):

484 articles scanned → 161 pairs aligned.

Critical source for medical term simplification.

Sozialpolitik.com:

22 articles → 22 pairs (100% success rate).

Educational content on the German social system.

Lebenshilfe Main-Taunus:

59 articles scanned → 46 pairs aligned.

Community-level communication and event reporting.

Global Scraping Statistics

Metric	Result
Total Scanned LS Articles	1,347
Successfully Aligned Pairs	797
Total Tokens (Simple)	411,540
Total Tokens (Standard)	676,903
Average Success Rate	~59%
Length Ratio (AS:LS)	1.65 : 1

Tokens: words and punctuation

Findings & Lessons Learned

Content Divergence: Standard articles are significantly longer, suggesting that simplification involves heavy summarization, not just sentence splitting.

Alignment Gaps: 41% of simple language articles lack a direct 1:1 standard counterpart on the same site.

Scale: With 797 pairs, the new corpus is already larger than most existing document-aligned German datasets.

Week 5

Focus: From Quantity to Quality

The Noise Problem: Raw scraping produced significant "boilerplate" (menus, ads, legal footers).

Implement source-specific minimal scripts to ensure that the tokens in the dataset are strictly editorial content (not menus, ads etc.).

Case Study: Apotheken Umschau

Issue: Figcaptions ("The image shows..."), internal table of contents (TOC), and "Sign up now" banners were polluting the text.

Solution:

Decomposition: Stripped <figcaption>, <figure>, and .copyright elements.

TOC Filter: Identified and removed blocks containing only internal anchor links.

Blacklist: Regex filters for "Das Bild zeigt" and registration prompts. (extend in future)

Case Study: Brand Eins

Issue: AS and LS content lived in the same HTML blocks; simple paragraph splitting failed due to irregular formatting.

Solution:

Deep-Color Inspection: Scraper now analyzes inline CSS for the red color codes (#ff0000) used by Brand Eins to highlight simple language.

Tag-Based Extraction: Using and span-styles as semantic markers for language levels rather than just structure.

Case Study: Hamburg.de

Issue: Many articles were machine-translated (MT), marked by a "Computer has translated this" disclaimer. These are unsuitable for a gold-standard corpus.

Solution:

MT-Detection: Automated scanning for MT-disclaimers; immediate exclusion of affected pairs.

Precision Alignment: Restricted search for AS-links to the official .km1-language-bar, reducing "Ghost Alignments" from 155 down to 57 high-quality pairs.

Case Study: MDR

Issue: "The Echo Effect" - Nested `<div\>` and `<p\>` tags caused the same text to be extracted twice.

Solution:

Parent-Check Algorithm: Elements are only extracted if none of their ancestors are already in the extraction list.

Idea: Length-Ratio Filter: Pairs with a ratio > 5.0 (e.g., a Live-Ticker vs. a short summary) are automatically discarded as "Unbalanced."

Case Study: Archival Sources

Cities of Cologne & Main-Taunus:

Encoding Issues: Resolved "Umlaut" errors (e.g., kÃ¶nnen) by forcing `response.apparent_encoding`.

Redundancy: Content hashing introduced to prevent duplicate articles caused by multiple URL parameters (m-20, m-79) pointing to the same archive snapshot.

Truncation: Automatically cutting off repetitive contact blocks ("Ansprechpartner:", "Telefon:").

Automated Data Cleaning Pipeline

HTML De-cluttering: Remove non-article tags (nav, aside, footer).

Structural Extraction: Map headers, lists, and paragraphs with proper spacing (\n).

Regex Filtering: Strip author credits, donation calls (taz), and radio-station promos (MDR).

Identity Filter: Compare LS and AS strings; discard pairs where the content is identical (common in news sites).

Deduplication: Hash-based check for uniqueness across the entire corpus.

Resulting Quality Improvement

Cleaner Data: Average noise reduction of 15-20% per article.

Better Alignment: Improved thematic coherence by filtering out "Hub-pages" and "Tag-overviews."

Final Count: A refined, manually verified collection of ~600-700 high-quality pairs (post-filtering).

Zwischenergebnis Corpus

Source	LS Tokens	AS Tokens	Total
apotheken	147,539	249,790	397,329
behindertenbeauftragter	27,822	34,113	61,935
brandeins	7,275	7,423	14,698
hamburg	44,358	40,063	84,421
koeln	40,868	24,182	65,050
main_tanus	7,289	6,715	14,004
mdr	70,761	100,608	171,369
sozialpolitik	6,654	14,651	21,305
taz	4,589	8,027	12,616
TOTAL	357,155	485,572	842,727

Summary of tokens (words and punctuation)

Next Steps

Apotheken Umschau: Investigation into why so few URLs from this source could be successfully aligned (root cause analysis for low alignment rate).

Length Ratios: Analysis of discrepancies between Plain Language (PL) and Everyday Language (EL) across various sources. Why do some sources systematically have more PL tokens than EL tokens, and others the reverse?

Historical data acquisition: Evaluation of the Wayback Machine to achieve broader coverage of articles from different time periods (past and present) and thus expand the corpus.

Template Script: Because of redundant code, build a template script with core functionalities

Week 6

Weekly Focus: City and State Portals

- **Strategy:** Systematic review of states and state capital portals (Stuttgart, Wiesbaden, Hannover).
- **Findings:**
 - High variance in the implementation of legal accessibility requirements.
 - Hannover identified as the absolute "top performer".
 - All sites have the same entrypoint page for "Leichte Sprache"

Legal Obligations

Why do many government sites offer so little parallel data?

1. **BITV 2.0 Minimum Solution:** The law only requires LS for:
 - Essential tasks (homepage)
 - Navigation & Accessibility statements
2. **No Full-Text Obligation:** News or technical articles do not legally have to be translated.

Case Study: Hannover.de (The Gold Standard)

- **Scale:** Over 800 article pairs – the largest single source in the corpus.
- **Technical Features:**
 - **Alignment:** URL logic - the AS version is primarily derived by removing the query parameter `?sp:out=easy` (or `?sp%3Aout=easy`) from the LS URL.
 - **Quality:** High coherence between Easy-to-Read (LS) and Standard Language (AS).

Technical Challenges: Surgical Cleaning

Problem: Standard extraction yields 10-20% boilerplate noise.

- *UI Fragments:* "Print page", "In sign language", "Send email".
- *Navigation:* "Back to overview", "Related topics".

Solution:

- Implementation of CSS blacklists (decomposition of containers like `.SP-Tools` , `.header-1s`).
- Regex-based filtering of repetitive standard sentences.
- **Result:** Significant increase in token precision and cleaner vocabulary statistics.

Problem: Wiesbaden Alignment

Insight: Technical alignment does not equal semantic alignment.

- **Symptom:** URL parameters suggest a translation, but content varies significantly.
- **Examples:**
 - AS: History of the city forest vs. LS: General forest rules.
 - AS: Press release on a poster campaign vs. LS: Explanation of "Smart City".
- **Action:** Introduction of a manual audit step before the model training phase. Wiesbaden data may be excluded.

Current Status (Metrics)

Source	Pairs	Words (LS)	Words (AS)	Tokens (LS)	Tokens (AS)
(...)	189	72,816	79,585	135,484	168,489
MDR (prev.)	235	53,869	83,487	98,780	173,765
Apotheken (prev.)	161	123,505	205,063	241,325	451,812
Hamburg (prev.)	57	34,124	33,688	61,455	73,137
Week 06 Additions					
Hannover	808	458,621	405,321	872,291	871,830
Stuttgart	42	23,653	46,060	45,202	106,629
Wiesbaden	41	7,138	10,127	13,808	23,332
Total	1,533	773,726	863,331	1,468,345	1,868,994

Tokens counted using the `tiktoken` library with the `cl100k_base` encoding.

Next Steps

1. **Finalize Extraction:** Complete final sources.
2. **(Ongoing):** Find more sources systematically.
3. **Train first model:** Train first model to see some first results.

Week 7

Weekly Focus: Measuring Information Loss

- **Hypothesis:** Texts in Easy Language (LS) don't just simplify syntax; they lose significant, concrete information compared to Standard Language (AS), even when they are longer due to explanations.
- **Goal:** Move beyond token counts and develop a multi-dimensional NLP methodology to quantify this information loss.
- **Corpus:** Analysis executed across the entire aligned dataset (1,501 article pairs).

Methodology: Similarity

1. Semantic Similarity (Embeddings)

- **Tool:** Sentence-BERT (`paraphrase-multilingual-MiniLM-L12-v2`).
- **How:** Calculate Cosine Similarity from AS and LS texts.
- **Why:** Captures whether the "core message" remains, even if heavily paraphrased.

Methodology

2. Fact Retention via NER Recall

- **Tool:** spaCy (de_core_news_lg).
- **How:** Extract entities (Persons, Locations, Organizations) from AS and check if they survive in LS.
- **Why:** Measures hard information loss. If names and places vanish, factual detail is lost or is being described differently.

3. Linguistic & Syntactic Metrics

- **Tool:** spaCy POS-Tagging.
- **How:** Analyze Lexical Density, Average Sentence Length, and Part-of-Speech (POS) shifts (e.g., ratio of Adjectives, Conjunctions).
- **Why:** Reveals *how* the simplification is achieved structurally (e.g., dropping conjunctions implies loss of subordinate clauses / hypotaxis).

Key Finding 1: Massive Loss of Factual Entities

Observation: Across almost all sources, the **NER Recall is extremely low (Ø ~15-25%)**.

- This means roughly 80% of specific entities (names, locations, dates) mentioned in the Standard Language text do *not* appear in the Easy Language version.
- **Conclusion:** LS achieves simplification primarily by omitting specific facts and generalizing content, confirming a massive factual information loss.

Key Finding 2: Token Expansion vs. Semantic Shift

Observation: In some sources like **Hannover** (Ratio: 1.89) and **Köln** (Ratio: 1.83), the LS texts are nearly twice as long as the AS texts.

- *Why?* Complex terms are replaced with lengthy, multi-sentence explanations.
- *Result:* Despite being longer, the semantic similarity drops to ~0.70. The text explains more, but delivers less original detail.

 Semantic Similarity

Key Finding 3: Structural Simplification (POS)

Observation: Clear shifts in word classes between complex and simple language.

- **Conjunctions & Adjectives:** Significant decrease in **simple** language.
- **Verbs vs. Nouns:** **Simple** language uses more Verbs and fewer Nouns.
- **Stylistic Shift:** This confirms a transition from **complex** to **simple** style, which is easier to process.

 POS Distribution

Summary of Analysis Results

Source	Token Ratio (LS/AS)	NER Recall	Sem. Similarity (SBERT)
apotheken	0.98	0.09	0.64
behindertenbeauftragter	0.91	0.30	0.75
hannover	1.89	0.25	0.71
koeln	1.83	0.20	0.68
mdr	0.85	0.22	0.73
stuttgart	0.97	0.24	0.90
sozialpolitik	0.46	0.10	0.71

Extract of the full analysis results across 1,501 pairs.

Mögliche Titelvarianten & Forschungsfokus

1. Allgemein & Flexibel:

- *Automatisierte Textvereinfachung für Leichte Sprache: Ein Framework zur datengestützten Modellierung und Evaluation*

2. Ausgewogen & Methodisch:

- *Neuronale Textvereinfachung in Leichte Sprache: Entwicklung domänenspezifischer Datensätze und automatisierter Bewertungsmetriken*

3. Spezifisch & Technisch:

- *Optimierung der maschinellen Übersetzung in Leichte Sprache: Aufbau eines Gold-Standard-Korpus und Training von Reward-Modellen zur qualitativen Modellsteuerung*

Week 8

Corpus Analysis

- **Problem:** Initial SBERT analysis (Week 7) was limited to **128 tokens**, resulting in >90% of articles being truncated.
- **Goal:** Increase model coverage and validate if semantic similarity holds across full-length articles.
- **Methodology Update:**
 - Scale up to **512 tokens** (MiniLM limit).
 - Introduction of **Jina Embeddings v2** (8192 tokens) for 100% coverage.
 - **Bidirectional NER:** Analysis of "Faktentreue" (LS -> AS).

 Verteilung der Artikellängen

Article Length Distribution

SBERT Limitation & Coverage Analysis

A statistical review of token counts revealed that the 128-token limit was capturing primarily "teasers" or "introductions," which often contain boilerplate or navigation instructions in LS.

Coverage	AS (Limit: 128)	LS (Limit: 128)	AS (Limit: 512)	LS (Limit: 512)
Fully Captured	9.4 %	5.1 %	56.9 %	52.9 %
Truncated	90.6 %	94.9 %	43.1 %	47.1 %

Lesson Learned: Even at 512 tokens, nearly half of the corpus is cut off. This necessitated the switch to a long-context model (Jina) to ensure total information retention during analysis.

Optimized Analysis: 128 vs. 512 Tokens

Increasing the context window significantly changed the measured similarity for most sources.

Source	Sim (128 Tokens)	Sim (512 Tokens)	Difference
apotheken	0.636	0.894	+0.258
hamburg	0.665	0.804	+0.139
koeln	0.684	0.833	+0.149
sozialpolitik	0.706	0.850	+0.144
wiesbaden	0.750	0.642	-0.108

Interpretation: For long texts (Apotheken, Köln), similarity rises with context. For others (Wiesbaden), it drops, indicating that while introductions are aligned, the core content diverges significantly.

Model Comparison: MiniLM vs. Jina (128 & 512 Tokens)

To ensure the model choice introduces no bias, we compared the original model (`MiniLM`) with the new one (`jina-embeddings-v2-base-de`) at identical limits.

Conclusion:

- **Same Trends:** Higher context equals higher similarity in both.
- **Stability:** Jina is more stable with longer texts (e.g., Wiesbaden doesn't drop abruptly). It validates that LS stays semantically close to AS.

Source	MiniLM (128)	Jina (128)	MiniLM (512)	Jina (512)
apotheken	0.636	0.688	0.894	0.800
behindertenbeauftragter	0.746	0.756	0.761	0.793
brandeins	0.637	0.549	0.698	0.599
hamburg	0.665	0.690	0.804	0.790
hannover	0.706	0.742	0.776	0.807
koeln	0.684	0.693	0.833	0.782
main_taunus	0.762	0.763	0.727	0.794
mdr	0.733	0.731	0.766	0.784
sozialpolitik	0.706	0.694	0.850	0.760
stuttgart	0.896	0.884	0.856	0.884
wiesbaden	0.750	0.754	0.642	0.777

Solving Truncation: The Jina Model (8192 Tokens)

Source	Jina (128 Tokens)	Jina (512 Tokens)	Jina (Full / 8192)
apotheken	0.688	0.800	0.836
behindertenbeauftragter	0.756	0.793	0.804
hannover	0.742	0.807	0.828
koeln	0.693	0.782	0.829
stuttgart	0.884	0.884	0.821

 Einfluss der Kontextlänge auf Semantische Ähnlichkeit

Solving Truncation: The Jina Model (8192 Tokens)

Bidirectional NER: Does LS "Invent" Facts?

We measured not just what survives (AS $\rightarrow$ LS), but also if LS introduces new entities (LS $\rightarrow$ AS).

Finding: Both metrics are low. LS doesn't necessarily invent facts, but it rephrases entities into common nouns (e.g., "Arbeitsagentur" $\rightarrow$ "Amt"), which current NER models fail to align.

 Bidirektionales NER

Correlation: Token-Ratio vs. Similarity

Finding: No significant linear correlation between text length expansion and semantic similarity.

Longer explanations in LS don't automatically guarantee higher semantic proximity to the original.

 Korrelation Token-Ratio vs Similarity

 Histogramm der Ähnlichkeitsverteilung

Manual Audit & Corpus Cleaning

Linguistic Shifts: Sentence Length

Analysis of 1,526 pairs confirms structural simplification.

- **Sentence Length:** Dropped from **15.6 tokens** (AS) to **9.1 tokens** (LS) – a 42% reduction.
- **Impact:** Drastic reduction in cognitive load per processing unit.

 Vergleich der Satzlängen

Next Steps

1. **Dataset Cleaning:** Apply the 0.6 - 0.98 similarity filter to create the "Gold Standard" training corpus. Also look for all other extremes like ratio too high etc.

Thesis-Titel

Entwicklung domänenspezifischer Datensätze und automatisierter Evaluation für ein Framework zur neuronalen Textvereinfachung in Leichte Sprache

Week 11

Lexical Diversity (Type-Token-Ratio)

Average Reduction: The lexical variety in LS is reduced by an average of **13.6 %** compared to AS.

- **Averages:** AS (0.778) vs. LS (0.672).
- **Top Simplifiers:** *Hannover* (0.656) and *Hamburg* (0.658) show the most rigorous vocabulary reduction.
- **Journalistic LS:** The *taz* (0.742) maintains the highest lexical diversity, indicating a "higher-level" Easy Language.

Visualization of Lexical Diversity

 MATTR Vergleich

Comparison of lexical diversity (MATTR) by source.

 TTR vs Length Scatter

TTR relative to text length (log-scale) with regression lines.

Week 12

Model Training Strategy

Goal: Establish a baseline for binary classification (Normal vs. Easy German) and identify optimal data filters.

- **First Step: Baseline:** Bi-directional LSTM (BiLSTM) for lightweight comparison.

Sentence-Level Classification

Similarity Range	Balanced Accuracy
0.60 - 0.98	92.48 %
0.70 - 0.98	92.43 %
0.80 - 0.98	92.99 %
0.90 - 0.98	90.55 %

The range 0.80 - 0.98 provides the best balance between data volume and semantic alignment.

Article-Level Classification

Does full context (up to 512 tokens) improve the distinction?

Similarity Range	Balanced Accuracy
0.60 - 0.98	95.93 %
0.70 - 0.98	97.30 %
0.80 - 0.98	99.03 %
0.90 - 0.98	98.44 %

Finding: The jump from **93 % (sentence)** to **99 % (article)** confirms that "Easy Language" is a holistic stylistic phenomenon that becomes nearly perfectly distinguishable at the document level.

Training Configuration (BiLSTM)

The following hyperparameters were used to achieve the baseline performance (99% BAcc):

Parameter	Value
Optimizer	AdamW
Learning Rate	10^{-3}
Weight Decay	0.01
Batch Size	32
Max Epochs	30
Early Stopping Patience	7
Dropout	0,4
Max Seq Len	512 (Articles)

Next Steps & Research Questions

1. **Out-of-Domain Validation:** Testing the models on external data (e.g., Lebenshilfe Kiel) to check for over-fitting on governmental styles.
2. **Hyperparameter Testing** Change hyperparameters systematically to find the best setup
3. **Long-Context Transformers:** Exploring models that can process up to 8,192 tokens to capture very long administrative articles without truncation.
4. **Vocabulary Pruning:** Investigating if removing rare words (< 3 occurrences) improves the classifier by forcing it to focus on common simplification patterns.

Week 13

Some Thoughts...

- **Paragraph Control Experiment:**

- filtered a paragraph-free version of the dataset (`lebenshilfe_dataset_no_paragraphs.json`).
- with `spaCy` we can filter out all whitespace tokens.

- **Vulnerability to Data Leakage Avoided:**

- Training data (governmental and news portals) and test data (internal documents, house rules, statutes from *Lebenshilfe*) are completely disjoint.
- An out-of-domain accuracy of >90% proves the BiLSTM captures genuine linguistic patterns of Easy Language rather than source-specific formatting.

- **Vocabulary Robustness:**

- Many specialized legal or administrative terms were absent from the training corpus and thus masked as `<unk>` (unknown tokens - freq ≤ 3).
- The model successfully classifies despite this, demonstrating reliance on structural features (sentence length, conjunctions, syntactic simplicity) rather than rote-memorizing vocabulary.

Content vs. Length Controls

Dummy Content Test

- All words replaced by a neutral token (▯), preserving original text lengths and padding patterns.
- **Balanced Accuracy: 50.0%** (classifies all as LS).

Constant-Length Slicing

- All texts sliced to exactly 50 or 100 tokens, eliminating all length variance.

 Comparison of balanced accuracy across scenarios

Weekly Focus: Out-of-Domain Evaluation (Lebenshilfe)

- **Objective:** Verify the generalization capability of the BiLSTM classifier on a completely independent, unpublished dataset from *Lebenshilfe*.
- **Source Data:** 98 unstructured text documents (`.docx` , `.doc` , `.odt` , `.rtf`) containing manual translations into Easy Language (LS) and their everyday language (AS) counterparts.
- **Result:** A cleaned JSON dataset with **49 verified AS-LS article pairs** (98 documents total).

Out-of-Domain Classification Performance

We evaluated the BiLSTM models (trained on the similarity sweet spot `0.80 – 0.98`) on the 98 *Lebenshilfe* texts without fine-tuning:

Metric	Article-Level Model	Sentence-Level Model (Aggregated)	Sentence-Level Model (Sentence-Level)
Balanced Accuracy	90.82%	97.96%	79.71%
LS Correct (Simple)	93.88% (46/49)	97.96% (48/49)	76.02% (5877/7731)
AS Correct (Normal)	87.76% (43/49)	97.96% (48/49)	83.41% (1961/2351)

Sentence-level aggregation is performed via majority voting on the sentences of each article.

Next Steps & Future Work

- **Regression:**

- *Concept:* Map the degree of simplification between standard (AS) and simple language (LS) texts.
- *Variant 1 (Mix-Up):* Create artificial mixed sentences of equal length from LS and AS to study step-wise transition boundaries.
- *Variant 2 (LLM-in-the-Loop):* Feed AS and LS text pairs into LLMs as input to predict and measure progression levels between different simplification stages.

Week 14

Weekly Focus: Implementing the Regression Model

- **Variant 1 (Mix-Up):** Create artificial mixed sentences of equal length from LS and AS to study step-wise transition boundaries.
- **Variant 2 (LLM-in-the-Loop):** Feed AS and LS text pairs into LLMs as input to predict and measure progression levels between different simplification stages.

Variant 2: LLM-based Generation of Synthetic Intermediate Steps

- **Objective:** Establish a pipeline for generating progressive intermediate steps of text simplification between everyday language (AS) and Easy Language (LS) using LLMs.
- **Implementation:** Created a dedicated text generation script.
 - **Inputs:** Paired articles from the Lebenshilfe dataset.
 - **Outputs:** Progressively simplified versions stored incrementally in a new JSON dataset.
 - **Configurable Steps:** Supported generation of any float steps between `0.0` (LS) and `1.0` (AS), defaulting to `0.25`, `0.50`, and `0.75`.
 - **Features:** Robust resume functionality (detects existing output and skips completed steps), input limit option, and post-processing/cleaning of LLM output.

Test Run & Lessons Learned (Ollama & Remote Endpoint)

- **Local Execution:** local execution using Ollama and Llama 3.
- **Remote Execution:** Server can be pinged successfully, but the model itself cannot be invoked.
- **Key Issues Identified:**
 - i. **Model Chatty Prefixes:** LLMs frequently prepend introductory text (e.g., *"Hier ist der Text..."*), necessitating an enhanced post-processing script logic.
 - ii. **Layout Loss:** Easy Language formatting (line breaks, bullets) was lost in lower intermediate steps (e.g., 0.25), requiring prompt refinement.

Implementation Design for Variant 1 (Mix-Up)

Approach B (Sentence-Level Mixup):

Build paragraphs mixing AS and LS sentences (e.g. 50% LS sentences, 50% AS sentences for step 0.50).

- **Question:** Which sentences should be selected to ensure the content still makes sense?

Week 15

Weekly Focus: Implementing & Evaluating the Regression Setup

- **Goal:** Realization of the regression approaches (Mix-Up and LLM-based synthetic steps) to predict continuous complexity scores $\lambda \in [0.0, 1.0]$.
- **Approach 1 (Sentence-Level Mix-Up):** Implementation of the sentence-blending logic (First Variant), generation of training paragraphs, and analysis of target distribution.
- **Approach 2 (LLM Step Generation):** Running step-wise text generation, resolving LLM formatting/prefix issues, and performing dataset alignment cleanup.

Approach 2: LLM Step Generation Execution

- **Pipeline Run:** Executed `generate_synthetic_regression_steps.py` on the remote GPU server (`FlensGen-GPT-05S-120B` model via VPN) and locally (LLaMA 3 via Ollama).

Remote Execution: Server can be pinged successfully, but the model itself cannot be invoked.

Approach 1: Mix-Up (First Variant)

First Variant Implementation:

- Sentence-level segmentation of parallel articles (LS & AS) using `spaCy`.
- Slices of randomly selected contiguous sentence ranges extracted independently from LS and AS and shuffled to build a paragraph.
- **Regression Target:** Calculated dynamically as the character length ratio of the LS portion to the total paragraph length.
- **Alternative Concept (Variant 2):** If the resulting target distribution of the first variant is not uniform enough for the regression model, a target $\lambda \sim U(0.0, 1.0)$ can be pre-sampled and sentence counts calculated accordingly.

Pseudocode (First Variant):

```
# Segment sentences
sents_ls = sentencize(ls_text)
sents_as = sentencize(as_text)

# Slice random contiguous blocks
start_ls, end_ls = rand_range(len(sents_ls))
start_as, end_as = rand_range(len(sents_as))
sample_ls = sents_ls[start_ls:end_ls]
sample_as = sents_as[start_as:end_as]

# Shuffle and calculate target
mixed = shuffle(sample_ls + sample_as)
target = char_len(sample_ls) / (
    char_len(sample_ls) + char_len(sample_as)
)
```

Target Distribution of the First Mix-Up Variant

- **First Variant Distribution:**

- The target distribution (peaking at 0.5 and near the boundaries) is usable for training, as the regression model should be robust enough.

- **Backup Concept (Variant 2):**

- Should the target distribution of the first variant lead to imprecise predictions at the extremes, Variant 2 is available as a backup concept where a uniformly distributed $\lambda \sim U(0.0, 1.0)$ is pre-sampled.

 First Variant Distribution

Next Steps & Research Questions

1. **Regression Training:** Implement model training based on the Mix-Up dataloaders.
2. **Remote Model Access:** Resolve the remote GPU server model invocation issue (server is pingable, but API does not return completions).

Week 16

Weekly Focus: Regression Training & Mix-Up Evaluation

- **Goal:** Train the regression model on the sentence-level Mix-Up approach.
- **Mix-Up Data Inspection:** Analyzed blended paragraph structure and manually checked calculated target values.
- **Model Stabilization:** Moved from an initial unstable setup (non-deterministic) to a fully converged, deterministic training pipeline.

Mix-Up Paragraph Generation & Target

- **DataLoader Logic:** Randomly extracts contiguous blocks from the Easy Language (LS) and Standard Language (AS) versions of an article, shuffles them, and computes the target.
- **Target (λ) Definition:** Character length ratio of the LS portion:

$$\lambda = \frac{\text{Length}(LS)}{\text{Length}(LS) + \text{Length}(AS)}$$

- **Linguistic Coherence:** Shuffling destroys logical paragraph coherence, which prevents the model from overfitting on semantic cues and forces it to focus strictly on sentence and stylistic complexity.

Mix-Up Paragraph Example

Source Sentences (German Extrakt):

- **Easy German (LS)** ($n = 2$):
 - *"Die Beauftragten der Bundes-Regierung für die Belange von Menschen mit Behinderungen..."*
 - *"Was macht der Behindertenbeauftragte..."*
- **Standard German (AS)** ($n = 5$):
 - *"Inhaltsverzeichnis Video: Was macht..."*
 - *"zum Download: Video: Was macht..."*
 - *"Gesetzlicher Auftrag..."*

Blended & Shuffled Paragraph:

Die Beauftragten der Bundes-Regierung... zum Download:
Video... Gesetzlicher Auftrag... Was macht der
Behindertenbeauftragte... Inhaltsverzeichnis Video...

Computed Regression Target (λ):

$$\lambda = \frac{\text{CharLen}(LS)}{\text{CharLen}(LS) + \text{CharLen}(AS)} \approx 0.2087$$

Initial Setup & Failure Mode (On-the-Fly Shuffling)

BiLSTM Regressor Setup:

- Embedding & Hidden Dim: 128, Dropout: 0.3, MSE Loss.
- Blending performed **on-the-fly** during `__getitem__`.

 Initial Scatterplot

Problem & Diagnostic:

- Val MSE: `0.0655`, Val MAE: `0.2099` (deceptive).
- **Diagnosis:** Shuffling on-the-fly made the validation set non-deterministic every epoch.
- **Prediction Collapse:** Lacking a persistent sequence structure, the model predicted the mean (≈ 0.45).

Solution: Deterministic & Pre-Generated Datasets

- **Pre-generation:** The mixed paragraphs are generated once during dataset initialization rather than on-the-fly.
- **Reproducibility:** Seeded random generator (42 for train, 99 for validation) ensures identical samples across all epochs.
- **Data Augmentation:**
 - **Train:** 10 fixed mixes per article pair → **9,280 samples**.
 - **Validation:** 2 fixed mixes per article pair → **208 samples**.

Final Results

Results at Epoch 23 (Early Stopping at 28):

- **Validation MSE: 0.0335** (halved from 0.0655)
- **Validation MAE: 0.1195** (halved from 0.2099)

Interpretation:

- **Continuous Curve:** Predictions align more with the diagonal $y = x$.
- **Extremes Accuracy:** Clean AS ($\lambda = 0$) and LS ($\lambda = 1$) are predicted with high confidence near their true values.

 Final Scatterplot

Next Steps

1. **Evaluate on Lebenshilfe dataset:** We should get only extremes?
2. **Evaluate LLM-Generated Levels (Approach 2):** Run the trained Mix-Up BiLSTM model on the synthetic LLM text stages (0.25 , 0.50 , 0.75) to assess if the predicted complexity aligns with prompt instructions.

Week 17

MixUp Regressors: Comparing the 4 Training Variants

- **Variant A: Static Pre-mixing**
 - *Concept:* Sentence mixing and shuffling are generated once during dataset initialization.
- **Variant B: Dynamic Mixing**
 - *Concept:* Random shuffling and mixing of sentences performed on-the-fly in `__getitem__`.
- **Variant C: Hybrid Solution**
 - *Concept:* Combination of static and dynamic. The probability of dynamic mixing $p_{dynamic}$ increases linearly across epochs from 0.0 to 1.0.
- **Variant D: Hybrid Solution + Cyclic LR**
 - *Concept:* Based on the Hybrid Solution (Variant C), but introduces a cyclic learning rate scheduler (`CosineAnnealingWarmRestarts`).

 Loss Curves Comparison

MixUp Regressors: Training & Validation Loss Curves

Evaluation on the Lebenshilfe Test Set

Model	$\emptyset \lambda$ (LS)	$\emptyset \lambda$ (AS)	Acc (0.5)	Balanced Acc	MAE (1/0)
A (Static)	0.6518	0.1176	87.16%	89.20%	0.2597
B (Dynamic)	0.5516	0.2811	77.41%	81.48%	0.3842
C (Hybrid)	0.6315	0.1323	83.18%	85.98%	0.2779
D (Hybrid + Cyclic)	0.7554	0.1051	91.78%	92.83%	0.1911

- **Best Results:** Variant D (Hybrid + Cyclic) dominates across all metrics.
- **Worst Results:** Variant B (Dynamic) struggles to learn stable representations.

 Targets Comparison

Analysis of the Lebenshilfe KDE Plots

Future Optimization Steps (Backlog)

- **Train with LLM-Generated Levels:**
 - Train the model on synthetic text stages (0.25 , 0.50 , 0.75).
- **Variant B with Cyclic LR:**
 - Train the purely dynamic mixing model (Variant B) using a cyclic learning rate scheduler (e.g., Cosine Annealing with Warm Restarts) to see if periodic momentum helps it escape local minima and overcome its initial convergence plateau.

Week 18

Weekly Focus: MixUp Optimization, Thesis Summary & Translation Model Planning

- **1. MixUp Regressor Optimization & Evaluation:** Cyclic LR evaluation on Variant B, diagnostic error analysis, and complete In-Domain & Out-of-Domain (Lebenshilfe) performance comparison across all 4 variants (MixUp & Non-MixUp).
- **2. Master Thesis Status Summary & Overall Architecture:** High-Level Overview for status quo.
- **3. Planning Translation Model (Step 3):** Model selection (Seq2Seq vs. Causal LLMs) and reward guided training.

Part 1: MixUp Regressor Optimization & Evaluation

1.1 MixUp Variant B + Cyclic LR: Loss & Scatterplot Analysis

 Loss Curve Variant B Cyclic

 Scatterplot Variant B Cyclic

1.2 In-Domain Evaluation (Test-Split)

A. Continuous MixUp Regression (MixUp Evaluation)

Model Variant	Test MSE	Test MAE
Variant A (Static)	0.0383	0.1557
Variant B (Dynamic)	0.0758	0.2264
Variant D (Hybrid + Cyclic)	0.0241	0.1027

B. Binary Classification on Pure Sentences (Non-MixUp Evaluation: LS = 1.0, AS = 0.0)

Model Variant	$\emptyset \lambda_{LS}$	$\emptyset \lambda_{AS}$	Accuracy (0.5)	Balanced Acc	MAE (Target 1/0)
Variant A (Static)	0.7596	0.2680	91.55%	91.46%	0.2526
Variant B (Dynamic)	0.6138	0.3312	80.17%	79.53%	0.3620
Variant D (Hybrid + Cyclic)	0.9007	0.1382	95.92%	95.93%	0.1164

1.3 Out-of-Domain Evaluation (Lebenshilfe Dataset)

A. Continuous MixUp Regression (MixUp Evaluation)

Model Variant	LH MSE	LH MAE
Variant A (Static)	0.0725	0.2111
Variant B (Dynamic)	0.0766	0.2212
Variant D (Hybrid + Cyclic)	0.0739	0.2087

B. Binary Classification on Pure Sentences (Non-MixUp Evaluation: LS = 1.0, AS = 0.0)

Model Variant	$\emptyset \lambda_{LS}$	$\emptyset \lambda_{AS}$	Accuracy (0.5)	Balanced Acc	MAE (Target 1/0)
Variant A (Static)	0.6533	0.1958	82.67%	85.75%	0.2888
Variant B (Dynamic)	0.6241	0.2766	87.42%	86.32%	0.3378
Variant D (Hybrid + Cyclic)	0.7293	0.0957	87.29%	89.37%	0.2036

1.4 In-Domain Test-Split Scatterplots: Non-MixUp vs. MixUp

Binary Classification (Non-MixUp: Pure Sentences):

 Test Classification Scatter

Continuous Regression (MixUp Blends):

 Test Regression Scatter

1.5 In-Domain Test-Split Density Distributions

Binary Classification (Non-MixUp: Pure Sentences):

 Test Target KDE

Continuous Regression (MixUp Blends):

 Test Regression KDE

1.6 Out-of-Domain Lebenshilfe Scatterplots: Non-MixUp vs. MixUp

Binary Classification (Non-MixUp: Pure Sentences):

 LH Classification Scatter

Continuous Regression (MixUp Blends):

 LH Regression Scatter

1.7 Out-of-Domain Lebenshilfe Density Distributions

Binary Classification (Non-MixUp: Pure Sentences):

 LH Target KDE

Continuous Regression (MixUp Blends):

 LH Regression KDE

Part 2: Thesis Status Summary & Big Picture

2.1 Big Picture: 3-Step Master Thesis Architecture

 Master Thesis Architecture

2.2 Step 1 Status: Data Corpus & Quality Assurance

Corpus Statistics (Final Cleaned State):

- **1,471** verified article pairs across 11 sources.
- **1.43M Easy German Tokens** vs. **1.78M Standard German Tokens**.
- **91,103 LS Sentences** vs. **54,256 AS Sentences**.

Quality Assurance Pipeline:

- **Long-Context Alignment:** `jina-embeddings-v2-base-de` (8,192 token window) for section-level matching.
- **Similarity Sweet-Spot:** Filtered range $0.80 \leq \text{Similarity} \leq 0.98$.

Top Sources	Pairs	Tokens (LS)	Tokens (AS)
Hannover.de	796	861,967	858,086
MDR	227	94,976	168,440
Apotheken Umschau	157	234,608	443,722
Hamburg.de	56	61,204	66,977
Behindertenbeauftragter	51	38,724	44,725
Wiesbaden.de	41	13,808	23,332
Lebenshilfe (Test)	34	9,974	11,037
TOTAL (Final)	1,471	1,429,433	1.781.714

2.3 Step 2 Status: Classification & MixUp Regression

1. Binary Classification Baseline Models

Model Architecture	Scope / Aggregation	In-Domain BAcc	Out-of-Domain BAcc (LH)
Sentence BiLSTM (Maj. Vote)	Sentence → Article	99.68%	96.94%
Article BiLSTM Baseline	Article (512 tokens)	99.03%	90.82%
Sentence BiLSTM (Raw)	Single Sentence	92.99%	78.76%

2. Continuous MixUp Regressor Variants

Model Variant	Training Strategy	Test MSE	Test MAE	In-Domain BAcc	Out-of-Domain BAcc (LH)
Variant A	Static (Pre-mixed)	0.0383	0.1557	91.46%	85.75%
Variant B	Dynamic (<code>__getitem__</code>)	0.0758	0.2264	79.53%	86.32%
Variant C	Hybrid Schedule	0.0267	0.1158	95.22%	85.10%
Variant D	Hybrid + Cyclic LR	0.0241	0.1027	95.93%	89.37%

2.4 Project Progress & Milestone Dashboard

Milestone	Status	Key Metrics & Achievements
1. Web Crawling & Corpus Building	Completed	1,471 Pairs, 1.43M Tokens, 11 Sources
2. Quality & Embedding Alignment	Completed	Jina-Embeddings-v2 (8k), Similarity 0.80–0.98
3. BiLSTM Classifier Baseline	Completed	99.68% Balanced Accuracy (Majority Vote)
4. Continuous MixUp Regressor	Completed	MSE 0.0241, MAE 0.1027 (Variant D)
5. Synthetic LLM Levels Eval	In Progress	Ollama / FlensGen-GPT Evaluation
7. Translation	Planning	mBART-50 & LLaMA-3-8B (LoRA/QLoRA) & Reward

Part 3: Planning Translation Model (Step 3)

3.1 Step 3: Model Architectures & Approaches

A. Sequence-to-Sequence (Encoder-Decoder)

- **Models:** `mBART-50` (`facebook/mbart-large-50`), `mt5` (`google/mt5-base` / `mt5-large`).

B. Causal LLMs (Decoder-Only via LoRA / QLoRA)

- **Models:** `LLaMA-3-8B-Instruct` , `Mistral-7B-v0.3` , `Qwen2.5-7B` .

Metric Models as Reward Function

- Reinforcement Learning / Direct Preference Optimization / Metric Guidance

Open Questions & Discussion

- **Reward Function:**
 - Proceed with a metric model (the best one) as the reward function?
- **Dataset Usage:**
 - Is it valid to train/evaluate multiple models (metric & translation models) on the same dataset?

Week 19

Weekly Focus: Thesis Structure, Evaluation of the Synthetic Regressor & SFT of the Translation Model

- **1. Master Thesis Structure:** Overall overview of Chapters 1-7.
- **2. Synthetic LLM-based Regression vs. MixUp:** Evaluation on the Lebenshilfe (LH) dataset, discussion on data diversity/volume, and analysis of context/input length (Jina context comparison & article length distribution).
- **3. Translation Model (Step 3):** Supervised Fine-Tuning (SFT) pipeline, visualization of the DPO architecture/workflow (Step 1 as pretrained mt5 removed) and DPO training progress.

Part 1: Master Thesis Structure Outline

1.1 Thesis Structure: Chapters 1-4

1. Einleitung

- 1.1 Problemstellung & Motivation (Barrierefreiheit)
- 1.2 Zielsetzung & Forschungsfragen
- 1.3 Aufbau (Drei-Stufen-Pipeline)

2. Stand der Forschung

- 2.1 Leichte Sprache (Regeln, Zielgruppen)
- 2.2 MT & Textvereinfachung (Seq2Seq, LLMs)
- 2.3 Evaluierungsansätze (Metriken, DPO)

3. Themenblock 1: Korpus-Erstellung

- 3.1 Multi-Quellen-Crawling (11-Quellen-Korpus)
- 3.2 Alignment & Das 1:n-Problem (Satz/Absatz)
- 3.3 Quality Assurance (Similarity Sweet-Spot)
- 3.4 Datenbereinigung & Normalisierung
- 3.5 Korpus-Statistiken (Diversität, TTR)

4. Themenblock 2: Metrik

- 4.1 Klassifikationsmodelle (BiLSTM vs. SBERT)
- 4.2 Out-of-Domain-Generalisierung & Bias
- 4.3 Continuous MixUp Regression
- 4.4 Evaluierung synthetischer Sprachstufen

1.2 Thesis Structure: Chapters 5-6

5. Themenblock 3: Übersetzung & Reward-Guided Fine-Tuning

- 5.1 Datenvorbereitung & Trainings-Splits (Gefilterter Sweet-Spot, Block-basiert)
- 5.2 Supervised Fine-Tuning (SFT) (mt5 / Mistral/LLaMA)
- 5.3 Reward-Guided Optimization (RLHF / DPO via MixUp-Regressor Reward)
- 5.4 Mehrdimensionale Evaluierung (Regelkonformität, NLI/NER Faktentreue)

6. Diskussion & Gesamtevaluation

- 6.1 Inhaltssynthese der drei Themenblöcke
- 6.2 Stärken, Grenzen & Systemrestriktionen
- 6.3 Beantwortung der Forschungsfragen

1.3 Thesis Structure: Chapter 7

7. Fazit & Ausblick

- 7.1 Zusammenfassung der Kernergebnisse
- 7.2 Ausblick & Zukünftige Arbeiten

Part 2: MixUp vs. Synthetic Regressor

2.1 Synthetic Data Generation & Model Training

- **Goal:** Generation of semantically consistent and grammatically fluent intermediate steps (0.25 , 0.50 , 0.75) with FlensGen-GPT-05S-120B .
- **Generated Datasets:**
 - **LH-Eval:** lebenshilfe_dataset_with_steps.json (245 samples)
 - **Main Corpus:** corpus_final_with_steps.json (7,380 samples)
- **Model Training:**
 - Trained BiLSTMRegressor on flattened text-target pairs.
 - Validation MSE reached a minimum of 0.0401 in epoch 9.

2.2 Learning Curve: Synthetic BiLSTM Regressor

 Learning Curve Synthetic

2.3 Systematic Comparison: MixUp vs. Synthetic LLM Model

Evaluation on the exact same test set (Lebenshilfe with 5 levels):

Metric	MixUp Model (Variant D)	Synthetic LLM Model	Gain from Synthetic Approach
MSE	0.1388	0.0786	-43.4 %
MAE	0.3079	0.1816	-41.0 %

Bias?

2.4 Boxplot & Regression Comparison

 Boxplot Comparison

2.5 Discussion: MixUp vs. Synthetic Data

- **Evaluation Fairness:**
 - Evaluating on the Lebenshilfe (LH) test set with synthetic steps is highly advantageous for the synthetic model. This is because this test set was generated using the same principle/process as the training data of the synthetic metric model.
- **Data Volume & Diversity:**
 - Does the synthetic approach lead to more data than MixUp through the intermediate steps?

2.6 Input Length Issues with Metric Models

- **The Problem:** In the metric models, the original articles are often longer than the maximum input limit of the embedding model.
- **Is this a critical issue?**
 - Probably **not**, since a complexity metric is primarily supposed to learn sentence structure and syntactic patterns, which are usually sufficiently constant and homogeneous across the entire article.
- **Influence of Context Length (Jina Embeddings):**
 - Tests with different context lengths for Jina Embeddings showed: while performance increases slightly with greater context length, it saturates early.

2.7 Visualization: Context Length & Length Distribution

Length Distribution of Articles:

 Article Length Distribution

Jina Context Comparison:

 Jina Context Comparison

Part 3: Translation Model: Seq2Seq SFT & DPO

3.4 DPO Workflow (Direct Preference Optimization)

- **Step 1 (Pre-training):**
 - Already completed in the mt5 base model and is not part of this work.
- **Step 2 (SFT):**
 - Supervised Fine-Tuning of the base model on aligned paragraph pairs.
- **Step 3 (DPO):**
 - Direct optimization of the model based on preference data, evaluated by the metric model as a reward.



Source: [Medium](#)

3.1 Pipeline Design & Training Steps

- **Base Model:** `google/mt5-small` (~300M parameters).
- **Training Data:** 1,471 aligned paragraph pairs (85% Train / 15% Val).
- **Out-of-Domain Test:** LH dataset.
- **Phase 1:** Supervised Fine-Tuning (SFT) using standard Cross-Entropy Loss.
- **Phases 2 & 3:** Candidate generation and evaluation via a combined (Composite) Reward:
$$R = 0.5 \cdot R_{\text{style}} + 0.5 \cdot R_{\text{sem}}$$
- **Phase 4:** Direct Preference Optimization (DPO).
- **Issue with input lengths?**

 Translation Pipeline

3.2 SFT Loss Curves (Overfitting Check)

- **Training Progress over 10 Epochs:**
 - Very steep drop in loss values in the first two epochs.
- **Validation Loss:**
 - **No Overfitting:** Validation loss values are not yet rising again, so more epochs could still be trained.

 SFT Loss Curves

3.3 DPO Training Status

- **Status:** DPO training is currently running. Results of the tuning and the final translation evaluation will follow in the next milestone.

Week 20

Weekly Focus: Metrics Bias, Master Corpus Consolidation, SFT vs. DPO Comparison & Web App Setup

- **1. Evaluation Metrics & Bias Analysis:** Length/boilerplate bias checks on simplicity scores, MixUp vs. Synthetic simplicity metrics, and similarity to traditional readability formulas.
- **2. Master Corpus Consolidation & Diagnostics:** Transition to a unified master pipeline, boilerplate cleaning analysis (before vs. after), and diagnostics insights.
- **3. SFT vs. DPO Comparison:** Metrics comparison table, visual analysis of evaluation, and qualitative translation sample comparison.
- **4. DPO Model Variant Comparison:** Comparison of different DPO tuning runs (trainer vs non-trainer, enriched datasets).
- **5. Web Application Setup:** Flask backend and Next.js frontend consolidation under web folder, model integration, and output simplification scale.

Part 1: Evaluation Metrics & Bias Analysis

1.1 Length Bias & Similarity Investigations

Length / Boilerplate Bias Check:

- **Goal:** Investigate if simplicity classifiers (reward models) possess a length bias (classifying shorter sentences as "simpler" regardless of semantic complexity).
- **Method:** Synthetically inflated sentences with empty boilerplate phrases.
- **Findings:** Some metric models show sensitivity to sequence length. Quantified this bias to prevent reward hacking during DPO alignment.

Metric Similarity & Correlation:

- **Goal:** Verify correlation between neural simplicity scores and classical readability indices (e.g., Flesch Reading Ease, Wiener Sachtextformel).
- **Findings:** Strong correlation found, but neural classifiers show significantly higher robustness for modern structures and idiomatic expressions.

1.2 Simplicity Metrics: MixUp vs. Synthetic Approach

- **Cross-Evaluation Setup:**

- Both simplicity metric models (trained via the MixUp approach and the Synthetic approach respectively) were cross-evaluated on both target datasets (LLM-generated levels and Sentence-MixUp) to analyze generalization and cross-validate their simplicity estimation capabilities.

- **Performance Characteristics:**

- The model trained with the **Synthetic Approach** performs exceptionally well on the LLM-generated levels (same domain as its training dataset).
- The model trained with the **MixUp Approach** exhibits superior robustness and lower error metrics when evaluated on the Sentence-MixUp dataset.

1.3 Evaluation on LLM-generated Levels

Metric	MixUp Approach (Variant D)	Synthetic Approach
MSE	0.1388	0.0786
MAE	0.3079	0.1816
Pearson r	0.6626	0.7412
Spearman rho	0.5885	0.7391

 LLM-generated Levels Comparison

1.4 Evaluation on Sentence-MixUp Dataset

Metric	MixUp Approach (Variant D)	Synthetic Approach
MSE	0.0892	0.1162
MAE	0.2302	0.2629
Pearson r	0.6939	0.5979
Spearman rho	0.7045	0.5917

 Sentence-MixUp Comparison

Part 2: Master Corpus Consolidation & Diagnostics

2.1 Unified Master Pipeline

- **Consolidation:**
 - Replaced fragmented scripts with a single unified master script.
- **Key Pipeline Features:**
 - **Data Alignment:** Loads cleaned sentence/paragraph pairs.
 - **Semantic Similarity:** Jina embeddings (8192 context window) via cosine similarity.
 - **Bi-directional NER Recall:** model for AS -> LS (info loss) and LS -> AS (hallucination checks).
- **Readability Scores:**
 - Integrates Flesch, Wiener Sachtextformel (WSTF), and LIX.
- **Lexical Diversity:**
 - Type-Token-Ratio (TTR) and Moving Average TTR (MATTR-50) to offset text length biases.
- **Information Loss & Outliers:**
 - Tracks Part-of-Speech (POS) shifts and detects outliers (noise filtering).

2.2 Boilerplate Analysis: Occurrence Before Cleaning

 Boilerplate Occurrence Before Cleaning

2.3 Boilerplate Analysis: Occurrence After Cleaning

 Boilerplate Occurrence After Cleaning

2.4 Corpus Diagnostics & Insights

NER & POS Analysis:

- Named entities are preserved in **over 80%** of simplified target sentences.
- Target texts (LS) exhibit significantly reduced lexical density (noun/verb distributions) and shorter sentences.
- Overall results indicate high alignment quality and clean datasets.

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Part 3: SFT vs. DPO Comparison

3.1 Metrics Comparison: SFT vs. DPO

Model	Ø Simplicity (R_style)	Ø Sem-Sim to AS (R_sem)	Ø Sem-Sim to LS Reference	Ø Composite Reward (0.5/0.5)
SFT Model (Base)	0.9061	0.8681	0.8368	0.8871
DPO Model (w_style=1.0, w_sem=0.0)	0.9345	0.8689	0.8383	0.9017

3.2 SFT vs. DPO Metrics Visualization

 SFT vs DPO Comparison

Part 4: DPO Model Variant Comparison

4.1 DPO Model Variant Comparison

Model Variant	Ø Simplicity (R_{style})	Ø Sem-Sim to AS (R_{sem})	Ø Sem-Sim to LS Reference	Ø Composite Reward (0.5/0.5)
1_dpo_w05_w05_final (Non-Trainer)	0.8422	0.8733	0.8334	0.8577
1_dpo_w05_w05_final_trainer (Trainer)	0.7491	0.8867	0.8418	0.8179
2_dpo_w10_w00_final (Non-Trainer)	0.9345	0.8689	0.8383	0.9017
2_dpo_w10_w00_final_trainer (Trainer)	0.6182	0.8938	0.8401	0.7560
3_dpo_w05_w05_enriched (Non-Trainer)	0.7769	0.9058	0.8551	0.8413
3_dpo_w05_w05_enriched_trainer (Trainer)	0.7774	0.8920	0.8370	0.8347

Part 5: Web Application Setup & Model Integration

5.1 System Restructuring & Frontend Integration

Technical Integration:

- Consolidated both frontend and backend directories under the unified web folder.
- **Backend:** Flask-API loads the translation model and exposes generation endpoints.
- **Frontend:** Built a modern Next.js interface providing real-time interactive translation utilities.

Interface Features:

- **Standardized Complexity Scale:** Translated outputs are mapped to a [0.0, 1.0] range (with 1.0 representing maximum simplicity).
- **Interactive Control:** Allows users to adjust generation parameters (e.g., beam search width, repetition penalty) dynamically.
- Dev server runs locally via standard execution commands for Flask backend and Next.js frontend.

Week 21

Weekly Focus: Pipeline Automation, Dataset Cleaning, LoRA SFT & Hallucination Mitigation

- **1. End-to-End Pipeline & HPC Automation:** Modular architecture, Slurm batch scheduling, and reproducible cluster workflows.
- **2. Advanced Dataset Refinement & Boilerplate Elimination:** Dedicated Lebenshilfe cleaner, deduplication, and surgical purge of navigation/SEO boilerplate.
- **3. LoRA / PEFT Integration for SFT:** Resolving severe full fine-tuning overfitting on mBART-50 (611M parameters) with ~0.25% trainable weights.
- **4. Decoding Parameter Calibration:** Systematic resolution of repetition loops vs. forced hallucinations via calibrated `repetition_penalty = 1.2`.
- **5. DPO Tokenizer Alignment & Native Training:** Resolving Seq2Seq padding issues via custom direct preference optimization loop.

Part 1: Pipeline Modularization & HPC Automation

1.1 Modular Pipeline Architecture & Slurm Jobs

Pipeline Modularization:

- Structured all codebase operations into 5 functional stages:
 - **Data Collection:** Crawlers and content scrapers.
 - **Preprocessing:** Filtering, normalization, glossaries, master corpus, and synthetic step generation.
 - **Evaluation:** Information loss, readability, and lexical diversity.
 - **Modeling:** Binary classifiers, regressors, SFT, and DPO.
 - **Visualization:** Comparison plots and reporting.

HPC / Slurm Integration:

- Standardized execution via dedicated batch job scripts.
- Optimized GPU resource allocation using partitioned 24GB MIG instances for efficient cluster scheduling.
- Established comprehensive cluster execution documentation for reproducible training runs.

Part 2: Advanced Dataset Refinement & Boilerplate Purge

2.1 Lebenshilfe Normalization & Duplicate Filtering

Lebenshilfe Dataset Cleaning:

- Implemented targeted filtering to remove non-editorial noise:
 - Auditor review tags ("*Prüfer*", "*testgelesen*").
 - Contact footers, signatures, and document metadata.
 - Normalized line-break punctuation (`\n.` → `.`) to preserve sentence boundaries.

Corpus-Wide Filtering:

- Filtered duplicate article pairs caused by URL parameter aliases (e.g., city archive redundancy).
- Eliminated extreme length ratio outliers (e.g., short summary vs. entire live ticker) to ensure balanced alignment.

2.2 Surgical Web-Boilerplate Elimination

The Problem: Recurring End-of-Text Prompts:

- Scraped texts contained recurring website navigation fragments:
 - **Apotheken:** *"Wo bekommen Sie noch mehr Informationen?"* (100% prevalence).
 - **Hannover:** *"Klicken Sie hier..."*, *"Wo finde ich weitere Infos?"* (>400 instances).
 - **Main-Taunus / BB:** *"Hier erfahren Sie mehr..."*, *"Sprechen Sie uns an!"*.
- Caused the translation model to hallucinate these phrases as closing boilerplate across all outputs.

Solution: Navigation Boilerplate Purge:

- Developed targeted regex and heuristic filter patterns:
 - Navigational question chains.
 - Teaser phrases and link references.
 - Interactive click invitations.
- Completely eliminated recurring boilerplate noise before model training.

2.3 Dataset Statistics Post-Boilerplate Cleaning

Metric	Raw Scraped Corpus	Cleaned Master Corpus	Delta / Reduction
Total Article Pairs	1,533	917	−616 (−40.2 %)
Simple German Words (LS)	773,375	390,922	−382,453 (−49.5 %)
Standard German Words (AS)	845,558	514,429	−331,129 (−39.2 %)
Boilerplate Prevalence	61.4 % of articles	0.0 %	Fully Eliminated

Part 3: SFT Model Optimization with LoRA (PEFT)

3.1 Overfitting Challenges in Full Fine-Tuning

Parameter Disparity:

- Base translation model: `facebook/mbart-large-50` (611M parameters).
- Cleaned dataset: 808 pairs (686 Train / 122 Validation).
- Full parameter fine-tuning led to extreme capacity disparity.

Observed Symptoms:

- Validation loss minimum at epoch 7 (1.94), diverging to 2.45 by epoch 15.
- Premature early stopping prevented learning complex syntactic simplification rules.
- Simplicity score median dropped from ~ 0.89 to 0.498.

3.2 LoRA Parameter-Efficient Fine-Tuning

LoRA Implementation:

- Low-Rank Adaptation freezes base weights and trains low-rank decomposition matrices:

$$\Delta W = B \cdot A, \quad A \in \mathbb{R}^{r \times d}, B \in \mathbb{R}^{k \times r}$$

- **Target Modules:** q_proj, v_proj, k_proj, out_proj, fc1, fc2.
- **Hyperparameters:** Rank $r = 16$, $\alpha = 32$, Dropout 0.05.
- **Trainable Parameters:** Reduced to **~1.5M** ($\approx 0.25\%$).

Training Dynamics & Stability:

- Enables higher learning rate ($LR = 1e-4$).
- Stable convergence over 30 epochs with patience = 10.
- Prevents memorization and enforces true syntactic simplification.
- **Automated Merge:** Merges weights post-training for seamless downstream compatibility with DPO and Web App.

Part 4: Decoding Parameters & Hallucination Mitigation

4.1 Empirical Analysis: Repetition Penalty Calibration

The Dilemma:

- Short input texts previously caused the model to either loop repetitive words or hallucinate unrelated topics at text endings.

Systematic Variations:

- **Penalty = 1.0:** Degenerative n-gram loops (repeating compound words endlessly).
- **Penalty = 2.5:** Over-penalizes core vocabulary; forces beam search into absurd topics (*"Katzen"*, *"Pflege-Heirat"*).
- **Penalty = 1.2: Optimal Sweet Spot** — fluid, accurate Leichte Sprache, stops cleanly with zero hallucinations.

Standardization:

- Calibrated `repetition_penalty = 1.2` across all inference and training setups:
 - Supervised Fine-Tuning (SFT).
 - Preference Dataset Generation.
 - Direct Preference Optimization (DPO).
 - Interactive Web Application.
 - Evaluation & Benchmarking Workflows.

4.2 Repetition Penalty: Qualitative Generation Sample

Input (Standard German):
"Hunde (Canis lupus familiaris) begleiten den Menschen seit Jahrtausenden und gelten als das älteste domestizierte Haustier. Sie stammen vom Wolf ab und haben sich im Laufe der Zeit perfekt an das Leben mit uns angepasst. Heute sind sie nicht nur beliebte Haustiere, sondern oft vollwertige Familienmitglieder, ausdauernde Arbeitstiere und therapeutische Helfer."

repetition_penalty	Generated Output (Simple German / LS)	Qualitative Assessment
1.0	<i>"...Hunde sind zum Beispiel: Wolfspudel. Wolfspudel ist ein Tier aus dem Wolfsgebiet. Wolfspudel ist ein Tier aus dem Wolfsgebiet..."</i>	Degenerative Loop: Endless n-gram repetitions.
2.5	<i>"Was sind Hunde? Hunde sind Haustiere... Wie kann ich meine Hunde adoptieren? ... Diese Arten heißen zum Beispiel: Katzen. Katzen sind besondere Haustiere... Pflege-Hilfe. Eine Pflege-Heirat..."</i>	Forced Hallucinations: Over-penalization forces beam search into absurd off-topic domains.
1.2	<i>"Was sind Hunde? Hunde sind Haustiere. Sie begleiten den Menschen schon seit Jahrtausenden. Hunde gehören zu den ältesten Haustieren. Sie stammen vom Wolf ab und haben sich im Laufe der Zeit perfekt an das Leben mit uns angepasst. Heute sind sie nicht nur beliebte Haustiere, sondern oft vollwertige Familienmitglieder, ausdauernde Arbeitstiere und therapeutische Helfer."</i>	Optimal Output: • Accurate simplification & explanatory flow. • Full semantic preservation. • Clean termination; zero hallucinations.

Part 5: DPO Tokenizer Alignment & Native Training

5.1 Tokenizer Alignment & Dedicated DPO Engine

Diagnosis of Framework Incompatibilities:

- Using third-party DPO trainers for `mBART-50` caused severe tokenization and padding discrepancies:
 - Padding token mismatch between encoder prompt and decoder response.
 - Incorrect label masking during log-likelihood computation.
- Diagnosed via dedicated tokenization alignment benchmarks.

Native PyTorch DPO Engine:

- Replaced external framework dependencies with a custom, transparent DPO training loop:

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

- Explicit loss masking on response tokens.
- Reliable convergence and memory-efficient execution on GPU cluster.

Plan: Expert Human Evaluation (Lebenshilfe Kiel)

Expert Collaboration Confirmed:

- A domain expert from **Lebenshilfe Kiel** has agreed to evaluate translation outputs and metric predictions within our project timeline.

Blinded Study Design:

- Construct an anonymized, mixed test set:
 - i. Standard German source texts (AS baseline).
 - ii. Model-generated translations (SFT & DPO variants).
 - iii. Professional human translations (Gold Reference).
- The expert evaluates samples blindly without knowing the underlying source/model.

Core Objectives & Analysis:

- **Translation Performance:** Benchmark model outputs directly against human reference translations.
- **Distinction / Turing Analysis:** Can the expert reliably distinguish model translations from human translations?
- **Metric Correlation:** Quantify correlation between human ratings and our automated scores (BERTScore & Neural Simplicity Metric).

Expert Evaluation: Criteria & Discussion

Proposed Evaluation Scale (0 – 10):

- *Harmonized with metric scale [0.0, 1.0].*

1. Content Preservation (Inhaltlicher Erhalt):

- Are core facts and semantics preserved accurately without distortions or hallucinations? (*Corresponds to BERTScore*)

2. Leichte Sprache Compliance (Regelkonformität):

- How well are Leichte Sprache rules applied (syntax, sentence length, vocabulary)? (*Corresponds to Simplicity Metric*)

Optional Criterion for Discussion:

3. Linguistic Fluency (Sprachliche Qualität):

- Is sentence structure grammatically sound and naturally fluent?
- *Question for supervisor:* Is a separate fluency rating necessary, or is it sufficiently covered by the pre-trained base model & rule compliance?

Open Questions & Discussion: Thesis Writing

- **Depth of Foundational Explanations (Theoretische Grundlagen):**
 - How much foundational knowledge should be explained upfront in the thesis?
 - Where is the right balance between standard NLP fundamentals (Transformer architectures, Seq2Seq, LoRA/PEFT, DPO) and domain-specific rules (Leichte Sprache guidelines)?
- **Structure & Placement of Research Questions (Forschungsfragen):**
 - Should all detailed sub-questions (e.g., corpus curation, metric modeling, LoRA stability, DPO reward alignment) be explicitly enumerated upfront in the introduction?
 - Or should the introduction focus on overarching research goals, with specific sub-questions introduced at the beginning of their respective chapters?

Week 22

Weekly Focus: Empirical Scaling Laws, Quantitative Rule Adherence, Factuality Benchmarking & Long-Context DPO

- **1. Quantitative Rule Adherence & Benchmark Evaluation:** 14 linguistic metrics across 12 corpus sources ($N = 1,439$) and SFT translation vs. Lebenshilfe gold standard.
- **2. Empirical Sample Complexity & Data Scaling Laws:** Systematic scaling study for both the continuous MixUp BiLSTM metric ($M = 20$ sweet spot) and mBART SFT ($N \approx 162$ phase transition).
- **3. Qualitative Error Analysis & Factual Consistency Benchmark:** Diagnosing 4 SFT failure modes, proving the SBERT bi-encoder blindspot & NER dilemma, and validating NLI sentence contradiction filtering ($N = 177$).
- **4. Context Window Expansion & Long-Context Jina Embeddings:** Scaling from 256 to 1000 tokens and replacing MPNet-128 with Jina-8192, dropping DPO sentence truncation to 5.4%.
- **5. DPO Loss Formulation & Composite Reward Balancing:** Mathematical proof of length bias in unweighted sum DPO vs. length-normalized mean DPO; discovering the Pareto optimum at 0.7 Style/0.3 Semantic.
- **6. Thesis Synthesis & Next Steps:** Optimal end-to-end pipeline configuration and timeline for expert human evaluation with Lebenshilfe Kiel.

Part 1: Quantitative Rule Adherence & Corpus Benchmarking

1.1 The Linguistic Rule Framework (14 Metrics)

Syntactic & Structural Rules:

- **Sentence Length (MSL):** Max 8–12 words per sentence; long sentence ratio (> 12 words).
- **Subordination Ratio:** Main sentences only (Parataxe); eliminating relative, causal, and conditional sub-clauses.
- **SVO Order:** Subject-initial sentence preference.

Grammatical Rules:

- **Passive Voice Elimination:** Replacing passive with active transitive verbs.
- **Genitive Elimination:** Substitution by dative prepositional phrases ("von").
- **Verbal vs. Nominal Style:** Suffix density (`-ung` , `-heit` , `-keit`) & Verb-to-Noun Ratio.

Lexical & Readability Dimensions:

- **Polysyllabic Words (≥ 3 syllables):** Pyphen hyphenation engine.
- **Compound Hyphenation:** Splitting German compound nouns with hyphens (`-`) or Mediopunkt (`·`).
- **Standard Readability:** Wiener Sachtextformel (WSTF, target ≤ 6.0) and German Flesch Reading Ease.
- **Semantic Faithfulness:** Dense cosine embedding similarity (Jina v2 Long Context).

Tooling:

- Implemented automated `LeichteSpracheRuleAuditor` in Spacy (`de_core_news_lg`), `textstat` , and `regex` .

1.2 Corpus Rule Adherence: Source-Level Discrepancies ($N = 1,439$)

Source	Domain	AS Sentence Length	LS Sentence Length	Sentence Reduction (%)	AS Passive / Sent	LS Passive / Sent	Passive Red. (%)	AS Genitive	LS Genitive	Genitive Red. (%)	AS WSTF
BrandEins	Journalism	17.16	7.17	-54.2 %	0.12	0.02	-34.7 %	0.10	0.01	-86.2 %	11.39
Hamburg	Municipality	13.63	6.92	-47.2 %	0.07	0.01	-60.4 %	0.08	0.01	-90.8 %	11.63
Hannover	Municipality	11.96	6.67	-41.4 %	0.05	0.00	-57.8 %	0.06	0.00	-86.5 %	11.18
Sozialpolitik	NGO / Association	13.09	7.73	-40.3 %	0.07	0.01	-84.9 %	0.07	0.01	-86.3 %	11.24
Apotheken	Health	11.77	6.94	-39.9 %	0.04	0.01	-48.4 %	0.05	0.03	-35.8 %	11.03
Behindertenbeauftr.	Federal Agency	15.52	8.97	-39.7 %	0.05	0.05	-8.6 %	0.10	0.03	-73.6 %	11.96
MDR	News / Broadcast	12.13	7.70	-35.1 %	0.07	0.06	-2.6 %	0.06	0.00	-92.1 %	10.25
Koeln	Municipality	12.38	7.83	-34.9 %	0.09	0.01	-77.0 %	0.10	0.01	-90.7 %	11.66
Wiesbaden	Municipality	11.33	8.22	-18.5 %	0.07	0.02	-49.2 %	0.07	0.03	-49.7 %	14.25

 Corpus Rule Adherence Side-by-Side

1.3b Corpus Comparison: Sentence Length, Genitives, Passives & WSTF

1.4 SFT Translation Model vs. Lebenshilfe Gold Standard ($N = 37$)

Rule Dimension	Metric	1. Standard German (AS)	2. SFT Model (500 Tok)	3. Lebenshilfe Gold	SFT Reduction / Transformation
Syntax	Mean Sentence Length (MSL)	12.50 words	7.71 words	6.79 words	-38.3 % Length Reduction
Syntax	Long Sentence Ratio (> 12 words)	42.4 %	11.2 %	3.3 %	-73.6 % Long Sentence Drop
Syntax	Subordination Ratio (Subclauses/Sent)	0.108	0.069	0.018	-36.2 % Subclause Drop
Grammar	Passive Voice Ratio (per Sent)	0.093	0.023	0.020	-75.4 % Passive Elimination
Grammar	Genitive Noun Ratio (per Noun)	0.092	0.021	0.010	-77.4 % Genitive Elimination
Grammar	Verb-to-Noun Ratio (Verbal Style)	0.406	0.614	0.567	+0.208 Increase (Active Style)
Lexicon	Polysyllable Ratio (≥ 3 Syllables)	33.0 %	23.1 %	23.4 %	-30.0 % (Matches Gold Exactly)
Lexicon	Compound Hyphenation Ratio	2.9 %	16.9 %	11.7 %	+14.0 % Over-Hyphenation
Readability	Wiener Sachtextformel (WSTF)	11.62	7.28	7.31	-37.3 % (Matches Gold)
Complexity	Phrase Structure Tree (PST) Depth	27.11	50.00	22.25	+82.10 % Reduction in Complexity

 Comprehensive Rule Adherence Dashboard

1.5 SFT Model vs. Gold Standard: 9-Dimension Rule Dashboard

1.6 Qualitative Samples: Positive Simplification vs. Severe Defects

Positive Simplification (Fluent SVO & Lists):

Source: "Pfarrer Schüller ist in der Flüchtlingsbetreuung engagiert – in seinem Pfarrgarten sollen Wohncontainer aufgestellt werden. Wenn man keine Räume hat, soll man welche schaffen..."

"Als Pfarrer arbeitet er in der Flüchtlings-Betreuung. Er sagt: Wenn es genug Räume gibt, dann können auch andere Pfarren helfen. Zum Beispiel: • In der Pfarr-Gemeinschaft, • im Pfarr-Garten, • und in den Wohn-Containern."

- Clean active voice, simple SVO structures, hyphenation, bullet list explication.

Severe Semantic Defects (Distortions & Prefix Bias):

Source 1: "...zu 16 Jahren und einem Monat Haft verurteilt. Costa Concordia vor Giglio gekentert..."

"Francesco Schettino ist der frühere Kapitän... Er wurde verhaftet. Und er war 16 Jahre alt. ...Dort gab es eine schwere Erdbeben-Erschütterung."

- **Prison term becomes age** (16 J. Haft → 16 Jahre alt); **hallucinates earthquake.**

Source 2 (NZZ Zürich): "Gewinn von 20,47 Mio. €..."

"Sachsen-Anhalt Die NZZ... Im Jahr 2047 hat die NZZ einen Gewinn von 22,2 Mio. Euro..."

- **MDR Prefix bias + profit turns into year 2047**

Part 2: Empirical Sample Complexity & Data Scaling Laws

2.1 BiLSTM MixUp Metric: Sample Complexity Scaling

Experimental Design:

- **Axis 1 (Synthetic MixUp Multiplier M):**
 $M \in \{2, 5, 10, 20, 40, 80\}$ on 480 base pairs.
- **Axis 2 (Real Base Pairs N):**
 $F \in \{10\%, 25\%, 50\%, 75\%, 100\%\}$ ($N = 48 \dots 480$) at fixed $M = 20$.

The $M = 20$ Phase Transition:

- Below $M = 20$, the metric fails continuous regression ($R^2 \approx 0.27\text{--}0.35$, Accuracy $\approx 74.5\%$).
- **At $M = 20$:** R^2 doubles to 0.678, MSE drops from 0.0497 $\rightarrow$ **0.0248**, and Accuracy surges to 85.08%.
- **Saturation beyond $M = 20$:** Quadrupling effort to $M = 80$ yields < 0.7 percentage points gain. $M = 20$ is the proven Pareto sweet spot.

Exp Run	Pairs (N)	Multiplier (M)	Test MAE $\downarrow$	Test R^2 $\uparrow$	Binary Acc $\uparrow$
scale_m2	480	2	0.1867	0.2671	74.00 %
scale_m5	480	5	0.1778	0.3340	74.50 %
scale_m10	480	10	0.1775	0.3553	74.50 %
scale_m20	480	20	0.1121	0.6777	85.08 %
scale_m40	480	40	0.1038	0.6903	86.42 %
scale_m80	480	80	0.1050	0.6976	86.83 %
scale_f010	48	20	0.2086	0.0903	68.25 %
scale_f025	120	20	0.1587	0.4673	79.17 %
scale_f050	240	20	0.1461	0.5379	81.92 %
scale_f075	360	20	0.1272	0.6208	83.17 %
scale_f100	480	20	0.1121	0.6777	85.08 %

2.2 MixUp Insight: Real Diversity vs. Synthetic Remixing

Direct Comparison (960 Samples / Epoch):

- **480 Base Articles $\times$ 2 Mixes:**
 $\text{MAE} = 0.1867$, $R^2 = 0.2671$, $\text{Acc} = 74.00\%$
- **48 Base Articles $\times$ 20 Mixes:**
 $\text{MAE} = 0.2086$, $R^2 = 0.0903$, $\text{Acc} = 68.25\%$

Scientific Takeaway:

- At identical batch capacity, a broader corpus of real human texts is **3 $\times$ more effective** ($R^2 = 0.267$ vs. 0.090) than repeatedly remixing small base data.
- Real lexical and syntactic diversity cannot be fully substituted by synthetic permutation.

Dynamic Resampling Coverage:

- 480 base pairs $\times M = 20$ yield 9,600 training pairs per epoch.
- With epoch-wise dynamic λ -resampling across 40 epochs, the BiLSTM encounters up to **384,000 distinct sentence mixtures**.
- Validates that the current dataset size is mathematically sufficient for robust regression signals.

2.3 mBART SFT: Neural Scaling Law & Sample Complexity

Fraction (F)	Train Pairs (N)	Val Loss ↓	Simplicity (R_{style}) ↑	Semantics (R_{sem})	LS Reference (Sim_{ref})	BLEU ↑	ROUGE-L ↑	Avg. Tokens	Truncation Rate (%)
10 %	65	2.6771	0.2773	0.9400	0.8708	0.0038	0.0940	102.0	83.78 %
25 %	162	2.5182	0.4669	0.9254	0.8732	0.0074	0.1225	135.5	78.38 %
50 %	323	2.3352	0.3798	0.9426	0.8830	0.0095	0.1329	154.9	78.38 %
75 %	484	2.2153	0.4439	0.9237	0.8857	0.0101	0.1323	149.9	64.86 %
100 %	646	2.0624	0.4688	0.9428	0.8798	0.0113	0.1395	163.1	64.86 %

Empirical Scaling Insights:

- Monotonic Power-Law Scaling:** Validation cross-entropy loss follows strict power-law decay ($2.677 \rightarrow 2.062$).
- Linguistic Phase Transition ($N \approx 162$):** At 10% ($N = 65$), mBART suffers from *shortcut copying* ($R_{\text{style}} = 0.277$). At 25% ($N = 162$), it achieves a breakthrough (+68.4% to 0.467), autonomously learning SVO syntax, bullet lists, and compound hyphenation.
- Simplicity Plateau vs. Robustness Growth:** Adding data beyond $N = 162$ does not force deeper syntactic simplification under Cross-Entropy loss ($R_{\text{style}} \approx 0.47$), but triples BLEU ($0.0038 \rightarrow 0.0113$) and drops truncation ($83.8\% \rightarrow 64.9\%$).

 mBART SFT Scaling Curves

2.3b mBART SFT Empirical Scaling Curves (6 Panels: Loss, Metrics, Stability & Compute)

Part 3: Qualitative Error Analysis & Factual Consistency Benchmark

3.1 Qualitative SFT Failure Modes ($N = 9,030$ Evaluated Outputs)

1. Numerical Slot Shifts & Hallucinations:

- *Source*: Former captain sentenced to 16 years prison.
SFT: "He was arrested and was 16 years old." (Prison term $\rightarrow$ Age).
- *Source*: 20.47 million Euro profit.
SFT: "In the year 2047, NZZ earned 22 million Euro." (Profit $\rightarrow$ Year).
- *Source*: Sports car manufacturer Ferrari on stock market.
SFT: "Ferrari is a famous race car driver." (Entity category confusion).

2. Corpus-Induced Prefix Artefacts:

- Unconditionally prefixing articles with "Sachsen-Anhalt" due to high MDR broadcast representation ($P(y_1 = \text{"Sachsen-Anhalt"}) \gg 0$).

3. Pseudo-Word Neologisms:

- Forced syllable hyphenation causes bizarre creations: "Bischwürden", "Schritt-Früchte", "Kriegs-Kraft-Material".

4. Token Budget Exhaustion:

- Verbose rhetorical question chains consume early token budget, cutting off sentences abruptly at the 256-token limit.

MixUp Metric Immunity:

- While autoregressive SFT learns directional text priors, the discriminative BiLSTM metric pools globally over all tokens. A single regional prefix has $< 1\%$ weight and zero correlation with label $\lambda = 1.0$.

3.2 Factual Consistency Benchmark ($N = 177$)

Evaluated on 4 balanced test slices: (1) Gold Positives, (2) Real Model Hallucinations, (3) Random Shuffle Negatives, (4) Targeted Minimal Perturbations (1-word negation/number shifts).

Metric	ROC-AUC (Overall)	Gold Positives ($N = 77$)	Real Hallucinations ($N = 40$)	Random Shuffle ($N = 30$)	Minimal Perturbations ($N = 30$)	Diagnostic Finding
SBERT Similarity (R_{sem})	0.7722	0.8877 ± 0.05	0.8687 ± 0.07	0.6523 ± 0.09	0.7934 ± 0.12	Bi-Encoder Blindspot: Detects topic shuffle (0.65), but blind to fact errors (0.87).
NLI $P(\text{Contradiction})$	0.6223	0.2530 ± 0.16	0.3101 ± 0.13	0.3123 ± 0.09	0.4091 ± 0.26	Sharp Contradiction Sensitivity: Heavily penalizes fine semantic inversions.
NLI Factuality ($P_e - P_c$)	0.5840	0.4547 ± 0.10	0.4452 ± 0.06	0.4461 ± 0.04	0.4027 ± 0.18	Diluted on document-level by neutral background sentences.
NER Jaccard Overlap	0.4900	0.1345 ± 0.11	0.2032 ± 0.12	0.0039 ± 0.01	0.6778 ± 0.25	The NER Dilemma: Worse than random (0.49). Penalizes valid LS simplification.
Number Check	0.4610	0.6343 ± 0.10	0.7015 ± 0.08	0.3003 ± 0.08	0.8167 ± 0.05	Deterministic complement;

3.3 Diagnostic Visualizations: Metric Distributions & ROC Curves

 Factualy 4-Panel Comparison

 Factualy ROC Curves

Part 5: DPO Loss Formulation & Composite Reward Balancing

5.1 Loss Aggregation: Unweighted Sum vs. Length-Normalized Mean

The Length-Bias Dilemma in Classical DPO (**sum**):

$$\log \pi(y \mid x) = \sum_{t=1}^{|y|} \log \pi(y_t \mid x, y_{<t})$$

- Because $\log p \leq 0$, every additional explanatory token accumulates negative log-likelihood.
- Penalizes long, didactically rich Leichte Sprache paraphrases and rewards premature truncations.

Length-Normalized Mean DPO (**mean**):

$$\overline{\log \pi(y \mid x)} = \frac{1}{|y|} \sum_{t=1}^{|y|} \log \pi(y_t \mid x, y_{<t})$$

- Optimizes average confidence per token, eliminating length exploitation.

Model Variant	Simplicity (R_{style})	Semantics (R_{sem})	Composite Reward	Truncation Rate (%)
SFT Baseline	0.6734	0.8528	0.7631	78.38 %
DPO Sum (Classic)	0.6616	0.8590	0.7603	72.97 %
DPO Mean (Length-Norm.)	0.6938	0.8595	0.7767	67.57 %

Empirical Verification:

- **DPO Mean** achieves the highest Simplicity (0.6938) and Composite Reward (0.7767).
- **DPO Sum** falls behind SFT on Simplicity (0.6616) due to token accumulation penalties.

5.2 Composite Reward Weighting: Simplicity vs. Semantics

$$R(x, y) = w_{\text{style}} \cdot R_{\text{style}}(y) + w_{\text{sem}} \cdot R_{\text{sem, norm}}(x, y)$$

Configuration	w_{style}	w_{sem}	Simplicity (R_{style})	Semantics (R_{sem})	LS Reference (Sim_{ref})	Composite (0.7/0.3)	BLEU ↑	Truncation Rate (%)
SFT Baseline	—	—	0.6445	0.8628	0.8503	0.7100	0.0069	75.68 %
dpo_w05_w05	0.50	0.50	0.6747	0.8585	0.8561	0.7298	0.0091	72.97 %
dpo_w07_w03	0.70	0.30	0.6877	0.8540	0.8554	0.7376	0.0085	81.08 %
dpo_w10_w00	1.00	0.00	0.6756	0.8613	0.8552	0.7313	0.0075	64.86 %

Scientific Insights:

- 1. **Pareto Optimum at 0.7 Style/0.3 Semantic:** Maximizes simplicity (0.6877) and composite reward (0.7376) while maintaining semantic fidelity (0.8540).
- 2. **Implicit DPO Regularization ($w_{\text{sem}} = 0.0$):** Even with zero semantic reward, semantic similarity does not collapse (0.8613) because the reference model π_{ref} acts as a regularizing anchor.
- 3. **Reproducibility:** dpo_w05_w05 replicates dpo_mean from the loss aggregation study to the exact decimal.

Part 6: Thesis Synthesis & Next Steps

6.2 Next Steps: Human Expert Evaluation (Lebenshilfe Kiel)

- **Expert Evaluation Study Execution:**
 - Blinded evaluation of randomized source texts, SFT/DPO model outputs, and human gold translations with domain experts from **Lebenshilfe Kiel**.
 - Harmonized 0–10 scoring on Content Preservation (Factuality) and Leichte Sprache Compliance (Rule Adherence).
- **Thesis Writing Roadmap:**
 - **Chapter 3 (Methodology):** Rule Auditor (14 metrics), continuous BiLSTM MixUp derivation, and length-normalized DPO formulation.
 - **Chapter 4 (Empirical Experiments):** Corpus analysis, scaling laws ($M = 20$ & $N = 162$), factuality benchmarking ($N = 177$), and long-context alignment.
 - **Chapter 5 (Evaluation & Expert Study):** Automated metric correlations vs. blind expert human ratings.

Discussion & Open Questions for the Supervisor

1. Factuality Metric vs. Scope of the Master's Thesis:

- **State of Research:** We have proven that the models successfully learn syntax, morphology, Leichte Sprache rules, and high semantic fidelity. However, an off-the-shelf composite metric that reliably stops fine-grained numerical/entity hallucinations without human-in-the-loop filtering is still an open research problem (SBERT blindspot & NER dilemma).
- **Core Question:** Given the strict master's thesis timeline, is it necessary to build a "flawless end product", or is the scientific proof of feasibility, scaling boundaries, and diagnostic limitation analysis already a complete academic contribution?

2. Focus on Core Contribution vs. Engineering Polish:

- Should the remaining thesis time strictly prioritize **writing the manuscript** (formalizing the 14-rule framework, scaling laws, and evaluation results) and conducting the **expert study with Lebenshilfe Kiel**?
- Or should additional engineering effort be allocated to train custom cross-encoder NLI reward models for hallucination elimination?
- **Thesis Framing:** How to best frame the trade-off between generative simplification capabilities and automated factuality evaluation in the discussion chapter?